Coral Reef Assessment Report: Baseline Ecological Survey Report in Mtwara, Tanzania

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Table of contents

List of Figures

1Introduction

2Methods

2.1 Surveyed Sites and Geographic Coordinates

The coral reef assessment in Mtwara was conducted across a total of 8 survey sites, strategically selected to represent a range of reef conditions and management regimes within the region. These sites include Chamba Cha Kati, Chamba Cha Kati, Kieti, Matenga, Mmaluko, Rasmo, Viulu, and one additional location. Each site was categorized based on its location relative to Marine Management Areas (MMAs), with some sites located inside MMAs and others outside. This selection allows for comparative analysis of ecological conditions under different management contexts, providing insights into the effectiveness of conservation efforts and the status of coral reef ecosystems in Mtwara.

Table 2.1 provides a clear baseline overview of the coral reef assessment sites in Mtwara, listing site names, reef names, management status, and geographic coordinates. The table orients readers to the spatial distribution of locations sampled for benthic substrate, invertebrates, fish biomass, and recruits, and clarifies whether each site is inside or outside established marine management areas. This contextual information supports interpretation of the subsequent ecological analyses across the study region.

Table 2.1: Geographic coordinates and management status of coral reef survey sites in Mtwara region.

			Coordinates	
Area	**Reef**	**Management**	**Latitude**	**Longitude**
MBREMP	Chamba Cha Kati		-10.31373	40.37449
MBREMP	Matenga		-10.32315	40.36270
MBREMP	Kieti	Core Zone	-10.27417	40.34190
MBREMP	Inner Membelwa		-10.27402	40.34183
MBREMP	Rasmo		-10.34772	40.38065
Mgao	Mmaluko		-10.10957	39.99297
Msanga Mkuu	Viulu	Open Access	-10.19407	40.21902
Msanga Mkuu	Nusura		-10.22044	40.21319

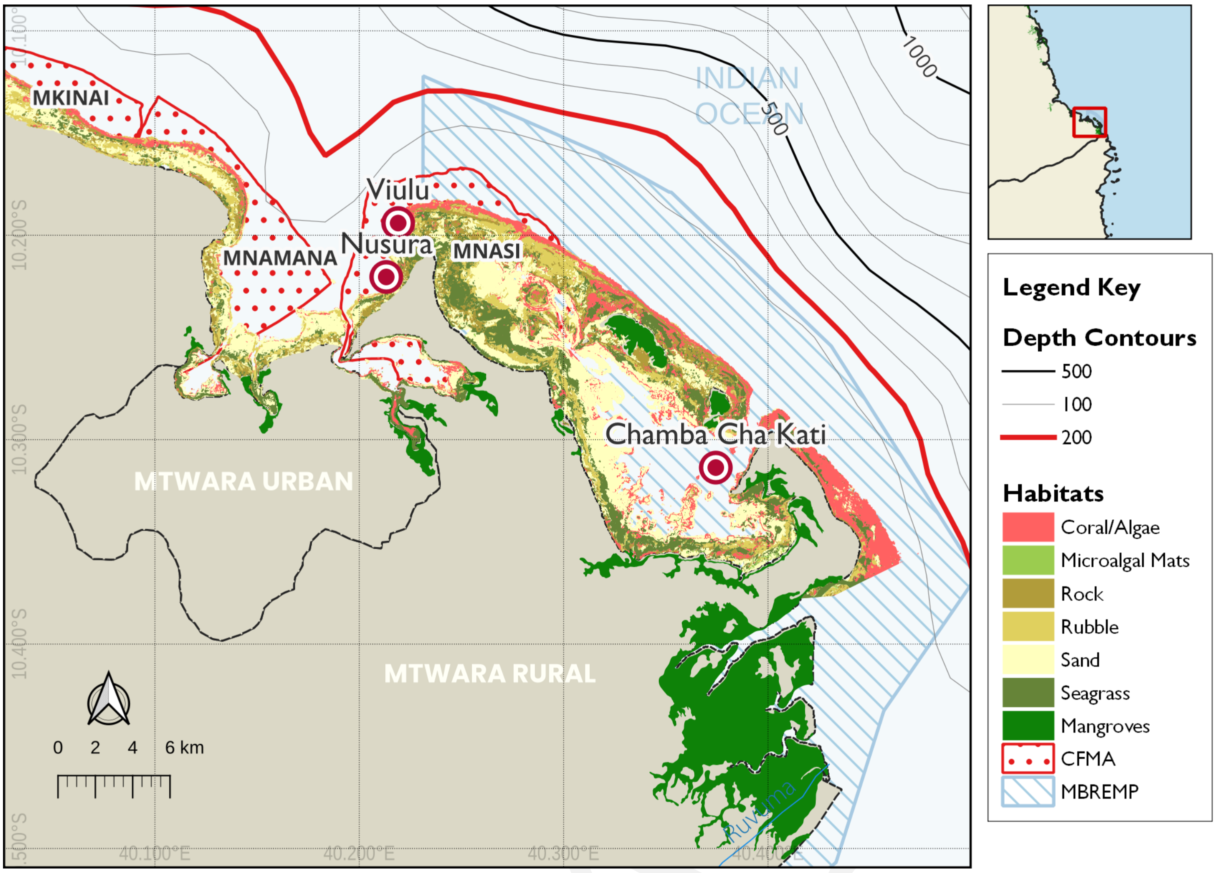

Figure 2.1: Distribution of survey site in Mtwara with spatial extent of the Mnazi Bay-Ruvuma Estuary Marine Park (MBREMP) showing the key benthic habitats zonation and proximity to the Ruvuma River mouth.

2.2 Survey Design

Ecological surveys were conducted between 16–24 March, 2026 following standardised reef health monitoring protocols (Hodgson et al. 2006; Wilson et al. 2020). Each site was assessed using replicated belt transects at two depth strata (shallow: 3–5 m; deep: 8–12 m), with three replicate transects per stratum per site (n = 6 transects per site). Site locations were recorded using GPS and mapped for repeatability in future monitoring rounds.

2.3 Data Processing

2.3.1 Benthic Substrate Assessment

Benthic cover was estimated using the Point-Intercept Transect (PIT) method (English, Wilkinson, and Baker 1997), where substrate categories were recorded at 0.5 m intervals along each 25 m transect tape, resulting in 50 data points per transect. These observations were classified according to the GCRMN scheme into seven primary categories: hard coral, soft coral, crustose coralline algae (CCA), algae (comprising both turf and macroalgae), rubble, sand, and other biota or substrate (e.g., rock, sponge). Finally, benthic cover estimates were derived by calculating the percentage of total points occupied by each category relative to the total number of points sampled per transect.

$$C_i = \left(\frac{n_i}{N}\right) \times 100 \tag{2.1}$$

where C_i is the percentage cover of substrate category i, n_i is the number of points where category i was recorded, and N is the total number of points sampled per transect. Equation 2.1 was used to calculate the relative abundance of benthic components.

2.3.2 Fish Biomass Assessment

Fish biomass and density were estimated using underwater visual census (UVC) belt transects (50 m x 5 m). All fish species observed within the belt were identified to family level, counted, and their total length estimated to the nearest centimetre. Fish density was calculated as the number of individuals per unit area (individuals per 100 m^2). Equation 2.2 was used to calculate the density of fish in an area.

$$D = \frac{N}{A} \times 100 \tag{2.2}$$

where D is density, N is the count of individuals, and A is the sampled area (250 m^2). Equation 2.2 describes the standard calculation for fish density.

Biomass (kg/ha) was calculated from length-weight relationships derived from FishBase (Froese and Pauly 2023), applying species-specific allometric parameters. Equation 2.3 was used to calculate the biomass of fish per area.

$$W = a \cdot L^b \tag{2.3}$$

where W is weight (g), L is total length (cm), and a and b are species-specific constants. Equation 2.3 is used to estimate individual fish weight for biomass calculations. Fish assemblages were characterised by trophic group (herbivores, piscivores, invertivores, and omnivores).

2.3.3 Coral Recruit Assessment

Coral recruit density was assessed using 1 m^2 quadrats placed at 1 m intervals along each transect (25 quadrats per transect). All coral recruits less than or equal to 5 cm maximum diameter were counted and identified to genus level where possible (Babcock et al. 1992). Recruit density is expressed as recruits 100 m^{-2} using Equation 2.4:

$$RD = \frac{\sum n_r}{A} \times 100 \tag{2.4}$$

where RD is the recruit density (individuals per 100 m^2), n_r is the number of recruits counted, and A is the total area sampled (m^2).

2.3.4 Invertebrate Assessment

Key indicator invertebrates were quantified using visual belt transects measuring 25 m in length and 2 m in width, covering a total area of 50 m^2 per replicate. The survey focused on ecologically and commercially significant taxa, specifically sea urchins (Diadema spp., Echinometra spp., and Tripneustes spp.) which serve as primary algal grazers; sea cucumbers (Holothuria spp., Thelenota spp., and Actinopyga spp.) which act as essential nutrient cyclers; and giant clams (Tridacna spp.) which are indicators of harvesting pressure. Abundance data were standardized to density per unit area to allow for cross-site comparisons. The density for each taxon was calculated using Equation 2.5:

$$ID = \frac{\sum n_i}{A} \times 100 \tag{2.5}$$

where ID represents the invertebrate density (individuals per 100 m^2), n_i is the total count of individuals for a specific taxon, and A is the total area surveyed (m^2). This standardization ensures that results are comparable with regional Western Indian Ocean (WIO) datasets.

2.4 Data Analysis, Visualization and Reporting

All data were analysed in R (v4.5.0; R Core Team (2024)). Site-level metrics are summarised as means +/- standard error (SE). Differences in reef metrics between were assessed using **rstatix** (Kassambara 2025). Figures were produced using the **ggplot2** (Wickham 2016) and **tidyplots** (Engler 2025), maps using the **sf** package (Pebesma 2018) and **QGIS** (Muenchow, Schratz, and Brenning 2017), while tables were formatted using **kableExtra** (Zhu 2024) and **gt** (Iannone et al. 2025). This report was prepared using **Quarto** within the **Positron IDE**.

3Results

3.1 Mtwara

3.1.1 Benthic Substrate Composition

The benthic community structure in Mtwara reflects a diverse coral reef ecosystem, with substrate composition varying significantly across the surveyed sites. At the site level, hard coral dominance was most pronounced at Chamba Cha Kati (80.8%), Matenga (80.6%), and Kieti (79.4%), as shown in Table 3.1. In contrast, Mmaluko exhibited a more heterogeneous composition with high coralline algae (43.9%) and sand (26.3%). Fleshy algae were most prevalent at Viulu (26.9%) and Mmaluko (23.8%), while seagrass was notably present only at Rasmo (22.5%) and Mmaluko (11.5%).

Table 3.1: Benthic substrate composition across survey locations in Mtwara

Substrate	Chamba Cha Kati	Membelwa	Matenga	Kieti	Rasmo	Viulu	Nusura	Mmaluko
Coralline	5.5	18.0	1.0	4.8	7.5	7.1	8.8	43.9
Fleshy algae	7.5	5.7	-	-	-	26.9	17.2	23.8
Hard Coral	80.8	41.5	80.6	79.4	63.5	54.9	65.8	43.9
Sand	3.0	13.3	6.6	10.6	9.3	7.6	12.8	26.3
Soft coral	1.9	2.1	2.3	8.4	7.0	13.1	6.5	-
Sponge	3.0	3.0	4.0	2.0	7.2	1.1	-	-
Algal Turf	-	33.0	6.1	14.0	-	-	-	12.6
Calcareous	-	6.5	-	-	-	-	-	-
Seagrass	-	-	-	-	22.5	-	-	11.5

Statistical analysis of the benthic substrate reveals significant differences in reef composition based on management status (Table 3.2). As illustrated in Figure 3.1, sites located within Marine Management Areas (MMAs) exhibit significantly higher hard coral cover compared to outside areas ($p < 0.001$), where algae and rubble dominate the landscape. Conversely, sites outside

MMAs are characterized by significantly higher levels of algae ($p = 0.001$) and rubble/other substrates ($p = 0.036$). No significant differences were observed for seagrass or soft coral cover between the two management regimes.

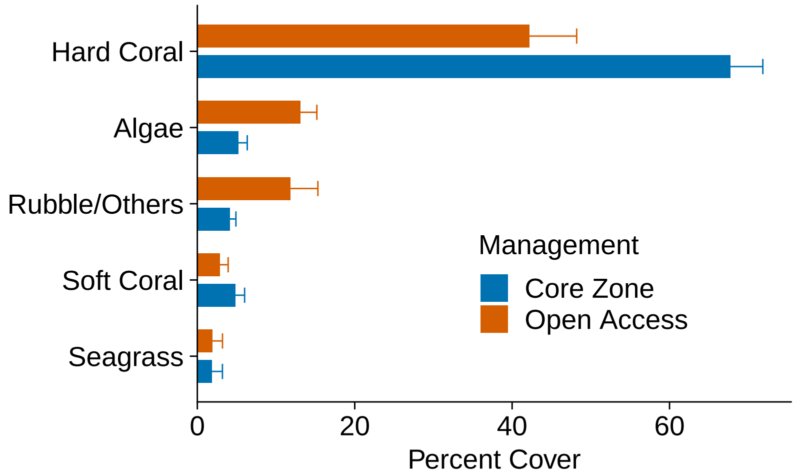

Figure 3.1: Benthic substrate composition across survey locations in Mtwara region, categorized by management status (Inside vs. Outside MMA).

Statistical analysis of the benthic substrate reveals significant differences in reef composition based on management status (Table 3.2). As illustrated in Figure 3.1, sites located within Marine Management Areas (MMAs) exhibit significantly higher hard coral cover compared to outside areas ($p < 0.001$), where algae and rubble dominate the landscape. Conversely, sites outside MMAs are characterized by significantly higher levels of algae ($p = 0.001$) and rubble/other substrates ($p = 0.036$). No significant differences were observed for seagrass or soft coral cover between the two management regimes.

Table 3.2: Results of t-tests comparing percent cover of major benthic substrate categories between sites inside vs. outside marine management areas (MMAs) in Mtwara region.

	Sample Size (N)				
Category	**Within**	**Outside**	**t-stat**	**p-value**	**Sig.**
Algae	108	70	-3.32	0.001	**
Hard Coral	30	24	3.51	0.001	**
Rubble/Others	53	33	-2.18	0.036	*

	Sample Size (N)				
Category	**Within**	**Outside**	**t-stat**	**p-value**	**Sig.**
Seagrass	24	12	-0.02	0.986	ns
Soft Coral	28	20	1.29	0.203	ns

3.1.2 Fish Biomass

Fish assemblages across Mtwara reefs showed pronounced spatial variability in density and family composition (Figure 3.2). **Chamba Cha Kati** (Mnazi Bay) recorded the highest fish densities overall, with Pomacentridae dominating at 254.6 individuals per 100 m^2 — nearly ten times the densities observed at **Nusura** (26.0) and **Viulu** (22.0) (Table 3.3). Herbivorous families, including Scaridae and Acanthuridae, were present across all surveyed sites, with Scaridae density peaking at Chamba Cha Kati (11.0 ind. 100 m^{-2}). **Viulu** stood out for comparatively higher densities of Siganidae (4.4) and Mullidae (5.0). Families such as Holocentridae and Balistidae were recorded exclusively at Chamba Cha Kati, while the unprotected reefs at Nusura and Viulu maintained higher densities of Lethrinidae and Pomacanthidae. These patterns suggest that although the protected reef at Chamba Cha Kati has greater overall fish biomass and damselfish abundance, the adjacent unprotected reefs continue to support functionally diverse assemblages that are important for broader reef resilience.

Table 3.3: Mean fish density (individuals per 100 m^2) by family across the surveyed reefs in Mtwara. Values represent the median density observed across all transects per site.

	Fish Density (individuals per 100m2)		
Family	**Chamba Cha Kati**	**Nusura**	**Viulu**
Pomacentridae	254.6	26.0	22.0
Scaridae	11.0	10.6	8.8
Labridae	7.4	11.0	8.2
Acanthuridae	5.0	3.6	3.4
Holocentridae	3.8	-	-
Chaetodontidae	3.6	1.0	1.6
Mullidae	1.6	3.6	5.0

	Fish Density (individuals per 100m2)		
Family	**Chamba Cha Kati**	**Nusura**	**Viulu**
Diodontidae	1.2	1.2	1.6
Balistidae	0.8	-	-
Siganidae	0.8	-	4.4
Fistularidae	0.4	-	-
Scorpaenidae	0.4	-	-
Lethrinidae	-	2.0	2.4
Lutjanidae/Nemipte	-	1.2	1.2
Others	-	3.2	-
Penguipedidae/Syn	-	1.0	0.6
Pomacanthidae	-	1.8	2.8
Serranidae	-	0.8	0.4
Pempheridae	-	-	1.2

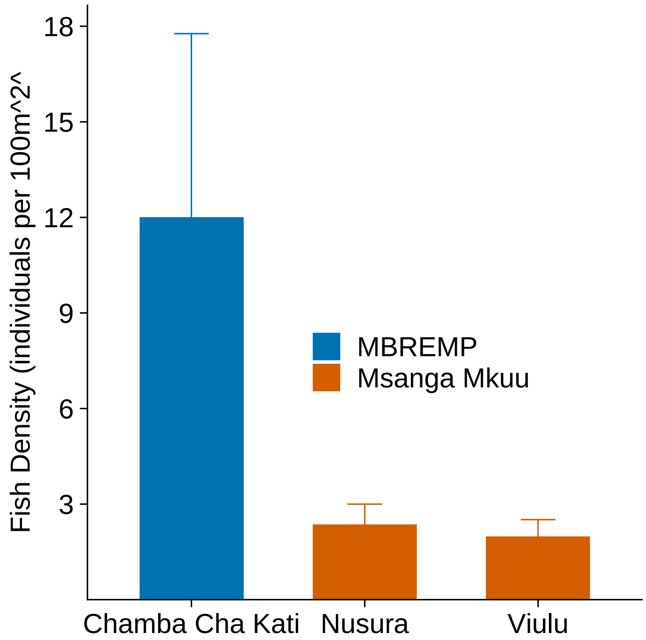

Figure 3.2: Median fish density (individuals per 100 m^2 +/- SE) across surveyed reefs in Mtwara. The dashed blue line represents the regional depletion threshold of 20 individuals per 100 m^2, indicating that most surveyed reefs currently sustain fish populations above critical depletion levels.

3.1.3 Coral Recruits

Coral recruitment across Mtwara reefs revealed considerable taxonomic diversity and spatial heterogeneity among the eight surveyed sites (Figure 3.3). At **Chamba Cha Kati**, recruitment was dominated by framework-building genera, with Acropora and Merulina each reaching densities of 500 ind. 100 m^{-2}, and Alveopora, Lobophyllia, and Millepora contributing equally at 300 ind. 100 m^{-2}. **Nusura** stood out for its exceptionally high Porites recruit density (650 ind. 100 m^{-2}), a pattern typical of reefs recovering from disturbance, where massive corals tend to dominate early successional stages. At **Viulu** and **Kieti**, recruitment was broadly distributed across Alveopora, Favia, and Stylophora (400–450 ind. 100 m^{-2}), suggesting relatively stable and diverse recruitment assemblages. The occurrence of specialist genera — Gardineroseris, Halomitra, and Herpolitha — exclusively at Chamba Cha Kati, alongside consistently elevated Acropora densities at **Kieti** (400 ind. 100 m^{-2}), points to meaningful recovery potential across the study region (Table 3.4).

Table 3.4: Recruit density (individuals per 100 m^2) by family across the surveyed reefs in Mtwara. Values represent the median density observed across all transects per site.

	Recruit Density (individuals per 100m2)							
Family	**Chamba Cha Kati**	**Membelwa**	**Matenga**	**Kieti**	**Rasmo**	**Viulu**	**Nusura**	**Mmaluko**
Acropora	500	100	100	400	300	100	200	250
merulina	500	-	-	-	-	-	-	-
Alveopora	300	-	-	-	-	400	100	-
Lobophyllia	300	100	100	-	100	100	100	100
Millepora	300	200	-	-	200	100	-	-
Porites(m)	300	-	500	100	200	450	650	500
Fungia	200	200	400	100	100	250	100	-
Gardineroser	200	-	-	-	-	-	-	-
Pavona	200	300	100	400	200	300	200	100
Plerogyra	200	-	250	-	-	-	-	-
Pocillopora	200	-	100	150	400	100	100	100
Favia	100	150	150	-	100	400	150	100
Favites	100	-	250	100	100	300	200	100
Galaxea	100	100	300	100	200	200	200	100
Halomitra	100	-	-	-	-	-	-	-
Herpolitha	100	-	-	-	-	-	-	-
Leptoseris	100	300	200	200	100	100	100	100
Platygyra	100	-	-	-	-	100	100	100
Seriatopora	100	150	100	150	200	-	-	-
Synarea	100	400	100	250	100	200	300	-
Psammocoral	-	300	-	-	-	100	100	-

Recruit Density (individuals per 100m2)								
Family	Chamba Cha Kati	Membelwa	Matenga	Kieti	Rasmo	Viulu	Nusura	Mmaluko
Stylophora	-	200	200	450	-	250	-	-
Echinopora	-	-	-	100	150	300	100	100
Merulina	-	-	-	100	100	-	-	-
Astreopora	-	-	100	-	-	400	100	100
Coscinaraea	-	-	100	-	100	-	-	-
Tubipora	-	-	100	-	-	-	100	-
Merullina	-	-	-	-	-	-	100	-
Pachyseris	-	-	-	-	-	-	100	-
Physogyra	-	-	-	-	-	-	200	-
Hydnophora	-	-	-	-	100	100	-	-
Acanthastrea	-	-	-	-	-	100	-	-
Psamocoral	-	-	-	-	-	100	-	-

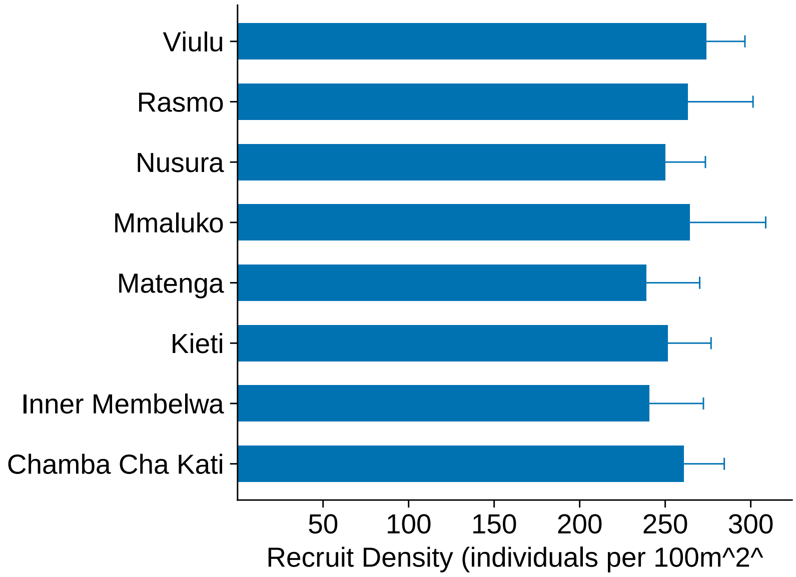

Figure 3.3: Median recruit density (individuals per 100 m^2 +/- SE) across surveyed reefs in Mtwara.

3.1.4 Invertebrate Assemblages

Indicator invertebrate densities in Mtwara varied significantly across the eight surveyed reefs (Table 3.5). **Viulu** exhibited the highest overall densities for several key groups, including sea stars (900 ind. 100 m^{-2}), urchins (1,150 ind. 100 m^{-2}), and gastropods (800 ind. 100 m^{-2}). High densities were also observed at **Rasmo** (650 sea stars) and **Kieti** (600 gastropods). **Nusura**, **Rasmo**, and **Mmaluko** supported the highest recorded densities of sea cucumbers (300 ind. 100 m^{-2}). In contrast, **Chamba Cha Kati** and **Membelwa** generally recorded lower invertebrate abundances, though Chamba Cha Kati maintained stable populations of bivalves (300 ind. 100 m^{-2}). Notably, Crown-of-Thorns starfish (COTs), a significant coral predator, were recorded at **Viulu** (200 ind. 100 m^{-2}) and **Kieti** (100 ind. 100 m^{-2}, See Figure 3.4), representing localized threats to reef recovery.

Table 3.5: Invertebrate density (individuals per 100 m^2) by family across the surveyed reefs in Mtwara.

Family	**Chamba Cha Kati**	**Membelwa**	**Matenga**	**Kieti**	**Rasmo**	**Viulu**	**Nusura**	**Mmaluko**
Sea star	400	200	300	400	650	900	600	650
Urchin	400	250	400	500	300	1,150	500	250

Family	Chamba Cha Kati	Membelwa	Matenga	Kieti	Rasmo	Viulu	Nusura	Mmaluko
Bivalves	300	200	200	400	350	500	350	-
Gastropods	250	400	300	600	300	800	400	400
Sea cucumber	150	200	200	300	300	200	300	300
Octopus	100	-	-	100	100	-	100	-
Worms	-	250	200	300	150	450	300	-
COTs	-	-	-	100	-	200	-	-

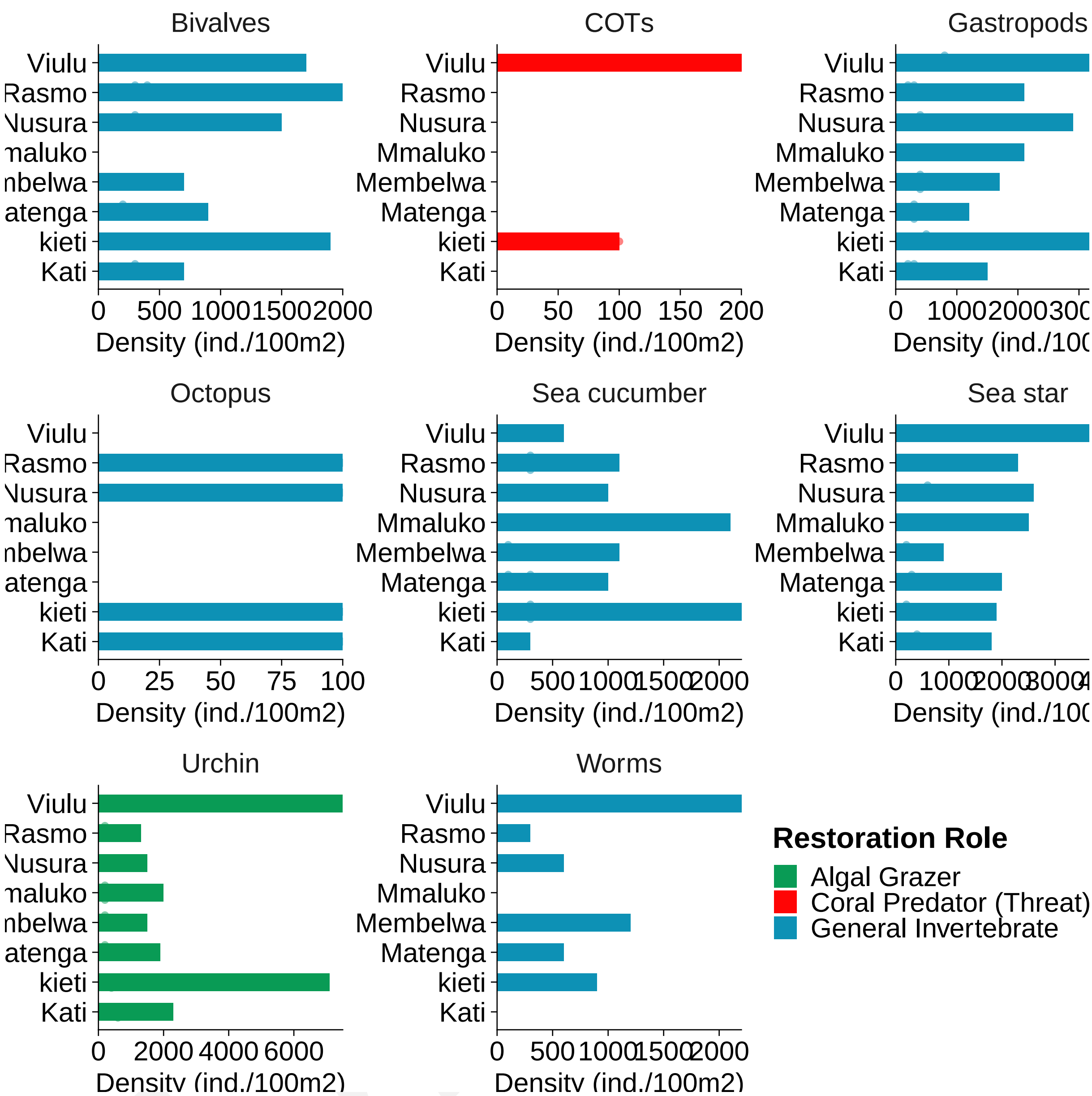

Figure 3.4: Invertebrate density (individuals per 100 m^2) by taxon across reefs in Mnazi Bay and Msangamkuu.

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Index